

ITA0606 – Machine Learning

Exp-11

Program:

Python Program for Credit Score Classification

Using Decision Tree Classifier

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

Step 1: Create sample dataset

```
data = {
```

```
    'Income': [25000, 30000, 45000, 50000, 60000,  
              70000, 80000, 35000, 40000, 90000],
```

```
    'Loan_Amount': [20000, 25000, 30000, 25000, 20000,  
                   15000, 10000, 30000, 25000, 10000],
```

```
    'Previous_Loans': [3, 4, 2, 2, 1,  
                      1, 0, 3, 2, 0],
```

```
    'Credit_Score': [  
        'Poor', 'Poor', 'Average', 'Average', 'Good',  
        'Good', 'Good', 'Poor', 'Average', 'Good'  
    ]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Step 2: Separate input features and target
```

```
X = df[['Income', 'Loan_Amount', 'Previous_Loans']]
```

```
y = df['Credit_Score']
```

```
# Step 3: Split the dataset into training and testing data
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
```

```
    test_size=0.3,
```

```
    random_state=42
```

```
)
```

```
# Step 4: Create Decision Tree model
```

```
model = DecisionTreeClassifier(
```

```
    random_state=42
```

```
)
```

```
# Step 5: Train the model
```

```
model.fit(X_train, y_train)
```

```
# Step 6: Predict credit scores
```

```
y_pred = model.predict(X_test)
```

```
# Step 7: Display actual and predicted values
```

```
print("Actual Credit Scores:")
```

```
print(y_test.values)
```

```
print("\nPredicted Credit Scores:")
```

```
print(y_pred)
```

```
# Step 8: Calculate accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
print("\nAccuracy:")
```

```
print(accuracy * 100, "%")
```

```
# Step 9: Display confusion matrix
```

```
cm = confusion_matrix(  
    y_test,  
    y_pred,  
    labels=['Poor', 'Average', 'Good']  
)
```

```
print("\nConfusion Matrix:")
```

```
print(cm)
```

Step 10: Predict credit score for a new customer

```
new_customer = [[55000, 20000, 1]]
```

```
prediction = model.predict(new_customer)
```

```
print("\nNew Customer Details:")
```

```
print("Income = 55000")
```

```
print("Loan Amount = 20000")
```

```
print("Previous Loans = 1")
```

```
print("\nPredicted Credit Score:")
```

```
print(prediction[0])
```

OUTPUT:

```
"" Actual Credit Scores:
['Average' 'Poor' 'Good']

Predicted Credit Scores:
['Average' 'Poor' 'Good']

Accuracy:
100.0 %

Confusion Matrix:
[[1 0 0]
 [0 1 0]
 [0 0 1]]

New Customer Details:
Income = 55000
Loan Amount = 20000
Previous Loans = 1

Predicted Credit Score:
Good
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but DecisionTreeClassifier was fitted with feature names
warnings.warn(
```